



Ginseng root extract attenuates inflammation by inhibiting the MAPK/NF- κ B signaling pathway and activating autophagy and p62-Nrf2-Keap1 signaling in vitro and in vivo

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ABSTRACT

Ethnopharmacological relevance: Panax ginseng C.A. Meyer is a type of herbal plant that has frequently been used in many Asian countries to treat a variety of diseases. Ginseng is considered to exhibit anti-inflammatory and anti-oxidative pharmacological effects. However, the specific mechanism is still not entirely understood.

Aim of the study: In this study, we investigated if ginseng extract could attenuate inflammation and oxidative stress in RAW264.7 cells and in dextran sulfate sodium (DSS)-induced colitis mouse model.

Materials and methods: RAW264.7 cells and LPS were used to develop inflammatory and oxidative cell models. C57/6J male mice and DSS were used to construct the animal models. O₂[•]-, mitochondria number, and mitochondrial membrane potential were analyzed using fluorescent probes and fluorescence microscopy. Reactive oxygen species and nitric oxide generation were detected with probes and microplate readers. The secreted amounts of inflammatory cytokines were measured by enzyme-linked immunosorbent assay kits. Protein expression levels in the cytoplasm and nucleus were measured by western blotting analyses. Quantitative real-time PCR (qRT-PCR) was used to determine the changes in mRNA levels. Autophagosome accumulation was analyzed by transmission electron microscopy. A p62-specific siRNA was used to evaluate the effect of p62 on the anti-oxidative function of ginseng root extract (GRE). Asperuloside and SP600125 were used to confirm the involvement of the MAPK/NF- κ B signaling pathway.

Results: We performed a systematic analysis of the anti-inflammatory, anti-oxidative, and autophagy regulatory mechanisms of GRE in LPS-treated RAW264.7 cells. GRE considerably reduced the levels of nitric oxide, TNF- α , and IL-6 secreted by LPS-treated cells. GRE treatments dose-dependently upregulated IL-10 mRNA levels and decreased IL-6 and IL-1 β mRNA levels in LPS-treated cells. Similar to the NF- κ B and JNK inhibitors, GRE treatment significantly inhibited NF- κ B activity and phosphorylation of MAPKs (JNK, ERK-1/2, and p38). Additionally, GRE treatment remarkably decreased LPS-triggered reactive oxygen species production and mitochondrial dysfunction by motivating Nrf2 nuclear translocation by enhancing phosphorylated p62. Knockdown of p62 resulted in the loss of GRE anti-oxidative ability. Autophagy was strongly induced by GRE via the Akt-mTOR signaling pathway, relieving excessive oxidation, mitochondrial dysfunction, and inflammation, while enhancing Beclin-1, LC3 II, and Atg7 protein expression. Furthermore, GRE alleviated the degree of injury, inflammatory cytokine production, and regulated the relative signaling pathway in DSS-induced colitis.

Conclusions: GRE can exert both anti-inflammatory and anti-oxidative functions by targeting the MAPK/NF- κ B and p62-Nrf2-Keap1 pathways, as well as autophagy, in vitro and in vivo.

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Table 1

Extract analysis.

	Low concentration (LC)	Middle concentration (MC)	High concentration (HC)
protein (μg/mL)	2.25	9	36
polysaccharide (μg/mL)	0.92	3.69	14.75
ginsenoside (μg/mL)	1.33	5.32	21.25

Table 2

Main components and contents of total ginsenosides.

Type	content (mg/g)	Total (mg/g)
R1	0.366	43.224
Rg ₁	5.383	
Re	7.304	
Rf	1.869	
Rg ₂	0.79	
Ro	0.971	
Rb ₁	6.176	
Rc	8.521	
Rb ₂	5.520	
Rb ₃	2.26	
Rd	4.032	
Rg ₃	0.029	
Rh ₂	0.003	

Table 3

Primers used for the qRT-PCR study.

Gene	Primer (5' to 3')
IL-6	F:CGGAGAGGAGACTTCACAGAG R:CATTTCACGATTCCAGAG
IL-1β	F:GCCACCTTTTGACAGTGATGAG R:ATGTGCTGCTGCGAGATTTG
IL-10	F:TAAGTGCACCCACTTCCAG R:AAGGCTTGCGCAACCAAGTA
GAPDH	F: AAGGTCATCCAGAGCTGAA R:CTGCTTCAACCTCTCTTGA

1. Introduction

Macrophages are known as sentinel cells in the innate immune system. These cells are located in the liver, lungs, kidneys, brain, skin, testes, intestines, vascular endothelium, and blood. Therefore, aberrant activation of macrophages can affect the outcome of different organ systems in a variety of diseases (Fujiwara and Kobayashi, 2005). Inflammation is an important protective response to pathological conditions through production of nitric oxide (NO) and pro-inflammatory cytokines. However, continuous activation of macrophages is detrimental, and this alone can induce the pathogenesis of diseases including arthritis, psoriasis, asthma, and cancer (Hotamisligil, 2006). Macrophages have core effects on the beginning and maintenance phases of inflammation, and dysregulated activation of macrophages that produce high levels of inflammatory factors have been implicated in various inflammatory diseases (Shapouri-Moghaddam et al., 2018). Hence, targeting aberrantly activated macrophages has become a key strategy for treating inflammatory disorders.

Inflammatory bowel disease (IBD) is a state of chronic intestinal inflammation and includes ulcerative colitis (UC) and Crohn's Disease (CD). The incidence of IBD is steadily increasing, with over 3 million people affected in the USA and an estimated 70 thousand new cases per year (Abraham and Cho, 2009). Accumulating evidence suggests that innate immune cells, mainly macrophages, play important roles in the development of IBD (Dharmasiri et al., 2021). Regulation of the balance between M1 and M2 macrophage phenotypes and oxidative stress were

believed to be crucial in the development of IBD, as well as possible targets for therapeutic intervention (Liu et al., 2014; Alzoghaibi, 2013). Reducing inflammation and oxidation are important strategies for treating IBD.

Anti-inflammatory mechanisms involve a variety of biological pathways. As a key transcriptional regulator, NF-κB can be activated by various cellular stimuli like certain cytokines, ultraviolet (UV) irradiation, oxidant free radicals, and pathogen products (Hayden and Ghosh, 2008). Activated NF-κB translocates to the cell nucleus and promotes the expression of genes with various biological functions. Increasing evidence shows that nuclear factor kappa B subunit 1 (NF-κB) has a considerable effect on the modulation of inflammatory factors such as NO, interleukin (IL)-1β, IL-6, cyclooxygenase (COX)-2, and tumor necrosis factor (TNF)-α (Afonina et al., 2017; Li et al., 2021). NO is produced by inducible nitric oxide synthase (iNOS) (Li et al., 2018). The anti-inflammatory cytokines with antagonistic effects include transforming growth factor (TGF)-β1, IL-4, and IL-10. These molecules negatively regulate the inflammatory response and reduce activity of the pro-inflammatory factors (Liu et al., 2014). Mitogen-activated protein kinases (MAPKs), including c-Jun NH2-terminal kinase (JNK), extracellular signal-related kinase (ERK) 1/2, and p38, are classic inflammation-related signaling molecules (Pearson et al., 2001). They are involved in regulating cellular responses to cytokine stimulation.

Additionally, oxygen metabolism plays an important role in the pathogenesis of inflammation. Reactive oxygen/nitrogen species (ROS/RNS) are involved in several inflammation-associated diseases. The accumulation of NO in turn induces increased levels of ROS/RNS (Brown and Borutaite, 2002). As the primary sources of intracellular ROS/RNS, mitochondrial dysfunction can be induced by excessive ROS/RNS. Nuclear factor erythroid 2-related factor 2 (Nrf2) is an important regulator in various antioxidant systems in cells responding to oxidative, inflammation, and metabolic stress and phototoxicity (Sho-keir et al., 2014). Nrf2 translocates to the nucleus and binds to antioxidant response elements (AREs). It can then induce a group of downstream targets related to cytoprotection, such as glutamate-cysteine ligase modifier subunit (GCLM), glutamate-cysteine ligase catalytic subunit (GCLC), and oxygenase 1 (HO-1) (Tebay et al., 2015). Kelch-like ECH-associated protein 1 (Keap1) is an Nrf2 inhibitor. Through the proteasome pathway, Keap1 targets Nrf2 for degradation and inhibits Nrf2 translocation to the nucleus (Kang et al., 2004). Some compounds can modify the cysteine residues of Keap1 that lead to conformational changes, resulting in stabilization of Nrf2 (Abiko et al., 2011). Additionally, Nrf2 stability is influenced by p62/SQSTM1 (especially serine 351 phosphorylated p62), which binds to Keap1 and moves it into autophagosomes, resulting in degradation of Keap1 and in turn stabilization of Nrf2 (Ichimura et al., 2013).

Autophagy, a kind of conservative cellular self-protection mechanism, is beneficial for degrading misfolded proteins and damaged mitochondria. Increasing evidence suggests that activation of autophagy can reduce ROS/RNS production, protect mitochondria, and decrease inflammation (Deretic et al., 2013; Levine et al., 2011). Different pathways participate in the regulatory relationship between autophagy and inflammation, such as the AKT serine/threonine kinase-mechanistic target of rapamycin kinase (AKT-mTOR), forkhead box O1 (FoxO1), and high mobility group box 1 (HMGB1) pathways (Wang et al., 2016; Zhu et al., 2015).

Panax ginseng C.A. Meyer is a type of herb that has been used across Asia for thousands of years. Ginseng has been widely used as a new resource food in recent years. The main pharmacological activities of ginseng include anti-inflammatory, antioxidant, anti-cancer, and immunoregulatory properties (Cambria et al., 2021; Hong et al., 2021). Ginseng can be used to treat various clinical diseases, and its protective effects on inflammatory reactions and oxidative stress are helpful for chronic diseases such as chronic fatigue syndrome, diabetes, neurodegenerative diseases, and others (Lu et al., 2021; Park et al., 2021). In this study, we used lipopolysaccharide (LPS)-induced RAW264.7 cells and a

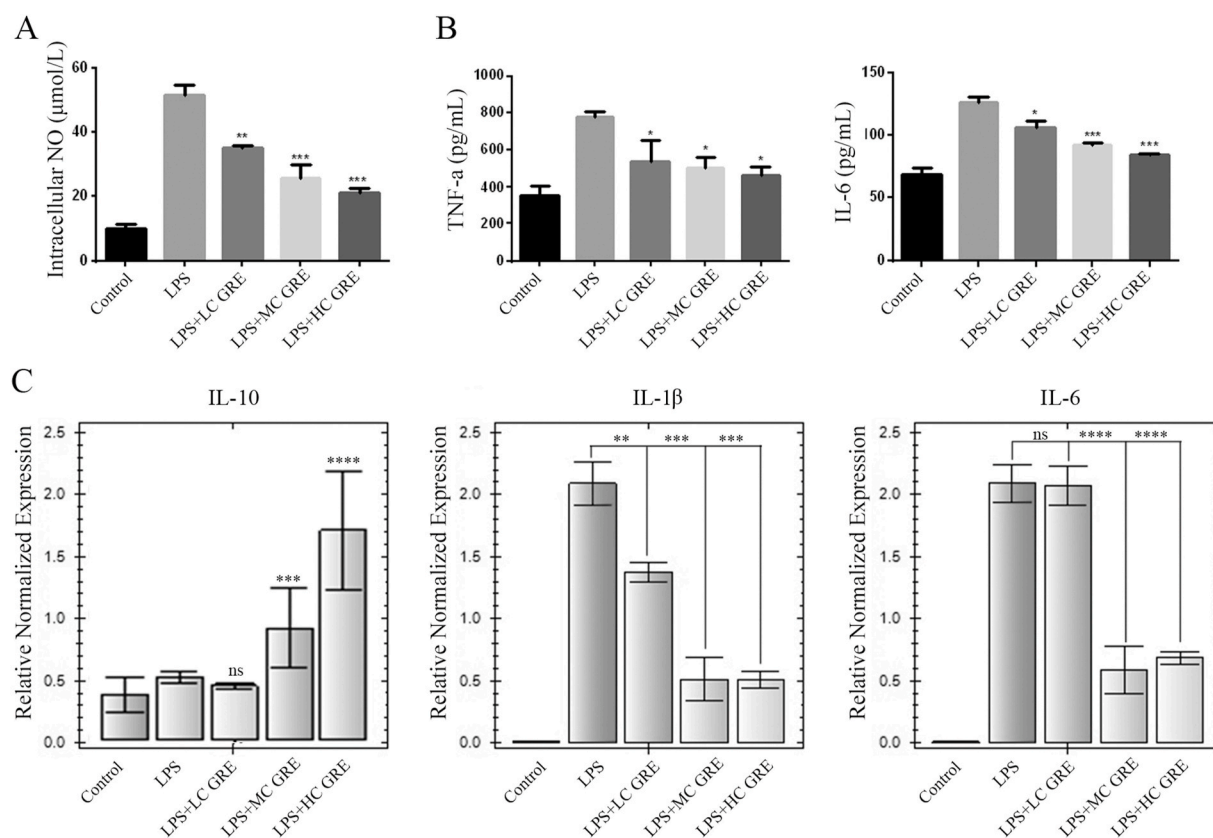


Fig. 1. Ginseng root extract (GRE) inhibits inflammation and production of pro-inflammatory factors triggered by LPS in RAW264.7 cells. (A) The NO level was measured by DAF-FM DA. The secreted levels of (B) TNF-α and (C) interleukin (IL)-6 were analyzed with ELISA kits. The mRNA levels of (D) IL-10, (E) IL-1β, and (F) IL-6 were detected by qRT-PCR.

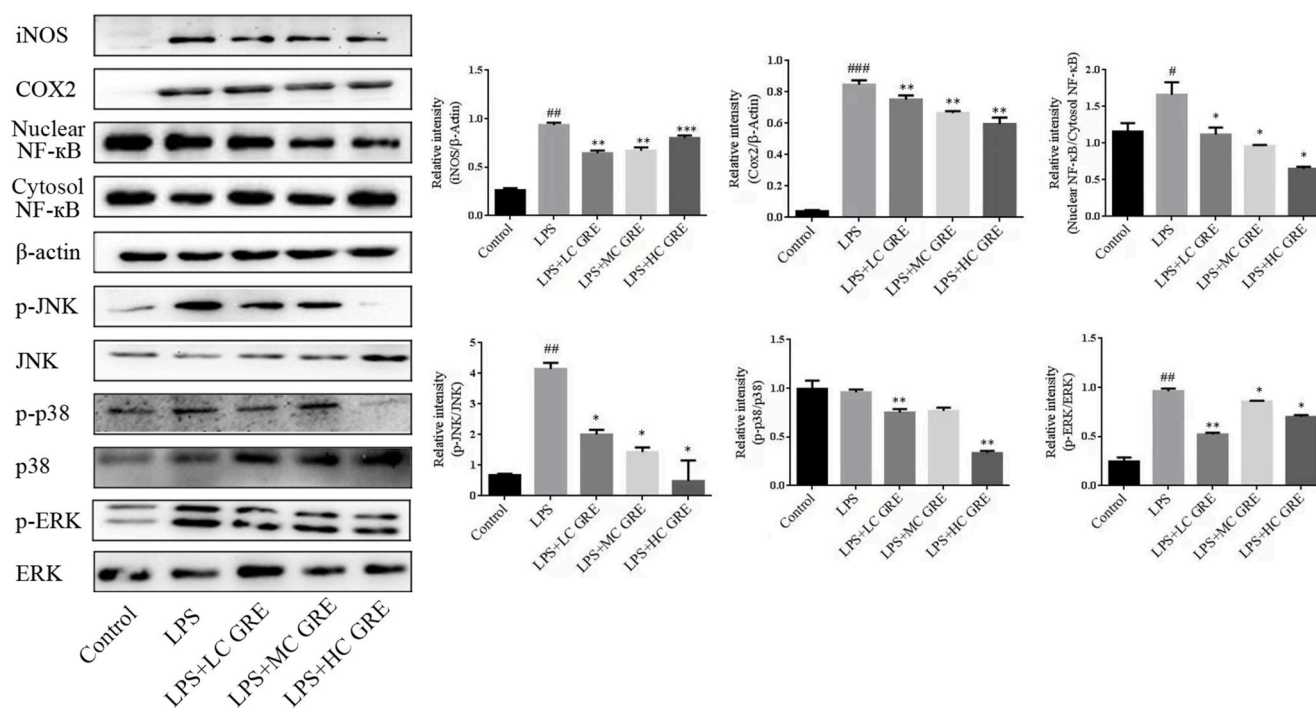


Fig. 2. Ginseng root extract (GRE) blocks LPS-triggered activation through the NF-κB and MAPK pathways. The protein expression levels of iNOS, COX-2, NF-κB, MAPKs, and β-actin were measured by western blotting using respective antibodies. β-actin was used as a protein loading control.

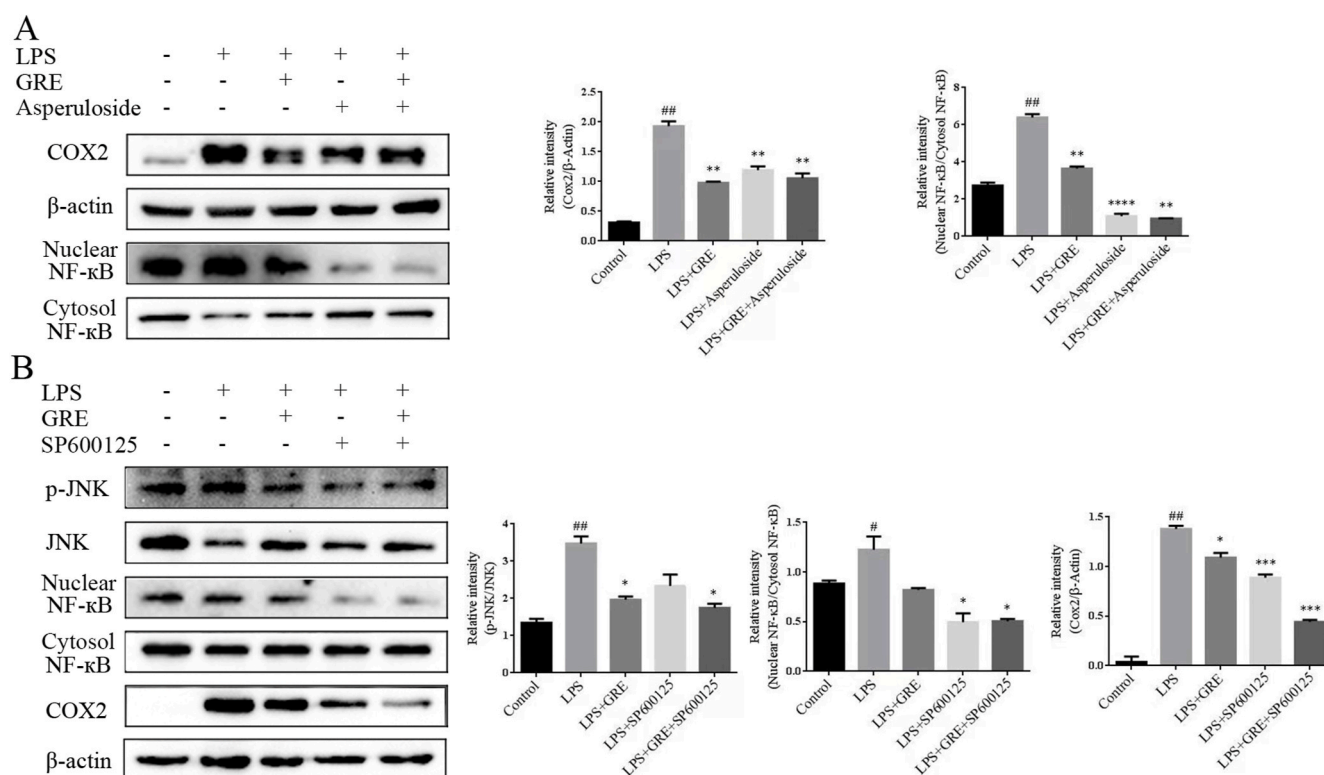


Fig. 3. Ginseng root extract (GRE) performs a similar function as MAPK and NF- κ B inhibitors. (A) After treated by asperuloside, the expression levels of COX-2, NF- κ B, and β -actin were measured by western blotting using respective antibodies. β -actin was used as a protein loading control. (B) After treated by SP600125, the expression levels of COX-2, NF- κ B, p-JNK, JNK and β -actin were measured by western blotting using respective antibodies. β -actin was used as a protein loading control.

DSS-induced colitis mouse model to evaluate any anti-inflammatory, anti-oxidative, and potentially protective effects of ginseng root extract (GRE). We also integrated the molecular mechanisms of its anti-inflammatory and anti-oxidative functions in these models.

2. Materials and Methods

2.1. Reagents and antibodies

The following primary antibodies were used in this study: antibodies against TAK1 (BS3745), Nrf2 (BS1258), p-ERK1/2 (BS4759), ERK1/2 (BS1968), p-p38 (BS4635), p38 (BS3566), p-JNK (BS4763), JNK (BS9825M), NF- κ B p65 (BS9879M), iNOS (BS90715), COX-2 (BS90326), GCLM (BS90567), p-AKT (BS4006), AKT (BS90043), p-mTOR (BS4706), mTOR (BS1555), and β -actin (AP0060) were purchased from Bioworld (Shanghai, China). Antibodies against p62 (5114), LC3 (12741), ATG7 (2631s), and Beclin-1 (3495s) were purchased from Cell Signaling Technology (Danvers, MA, USA). An antibody against GCLC (ab53179) was purchased from Abcam (Cambridge, UK). An antibody against p-p62 (M217-3) was purchased from Medical and Biological Laboratories (Woburn, MA, USA). An antibody against Keap1 (bs-3648R) was purchased from Bioss (Beijing, China). A horseradish peroxidase (HRP)-conjugated goat anti-mouse IgG secondary antibody (115-035-003) and an HRP-conjugated goat anti-rabbit IgG secondary antibody (111-585-003) were purchased from Jackson ImmunoResearch (West Grove, PA, USA).

The NO assay, dihydroethidium (DHE), Mito-Tracker, DCFH-DA, JC-1, lysis buffer, ECL western blotting system, BCA assay kit, and nuclear protein extraction kit were purchased from Beyotime (Shanghai, China). Enzyme-linked immunosorbent assay (ELISA) kits were purchased from Sino Best Biological Technology Co. (Shanghai, China). LPS was purchased from Yuanye Biotechnology Co. (Shanghai, China). SYBR Premix

Ex Taq™ and a Reverse transcription kit were purchased from TAKARA (Dalian, China). Advanced DNA RNA transfection reagent (AD600025) was purchased from ZETA LIFE (Menlo Park, CA, USA). Asperuloside (NF- κ B inhibitor) and SP600125 (JNK inhibitor) were purchased from MedChemExpress (Shanghai, China).

2.2. Ginseng extraction

Dry ginseng roots were purchased from Fusong, Jilin province, China and collected and identified according to the Chinese Pharmacopoeia (2015). The dry ginseng roots (50 g) were crushed and screened (80 mesh) at a solid-liquid ratio of 1:10, followed by ultrasonic extraction in a 40 °C water bath for 2 h, centrifugal filtration of the extraction solution, and ultrafiltration concentration. The residue was added with water material at a liquid ratio of 1:10 and extracted at 80 °C for 1.5 h. The extracted solution was concentrated by ultrafiltration. The concentrated solution was respectively freeze-dried and reserved.

2.3. GRE composition analysis

The polysaccharide content of the GRE was determined by the phenol-concentrated sulfuric acid standard curve method (Rasouli et al., 2014) after water extraction and alcohol precipitation. The content of saponin in the GRE was determined by spectrophotometry. The protein content of the GRE was determined using a BCA protein assay kit.

2.4. Analysis of total ginsenosides by HPLC (high performance liquid chromatography)

To analyze ginsenoside monomers, we used an Agilent 1260 series high performance liquid chromatograph, Agilent system chemistry workstation, and Agilent 1260 UV-visible wavelength detector. A

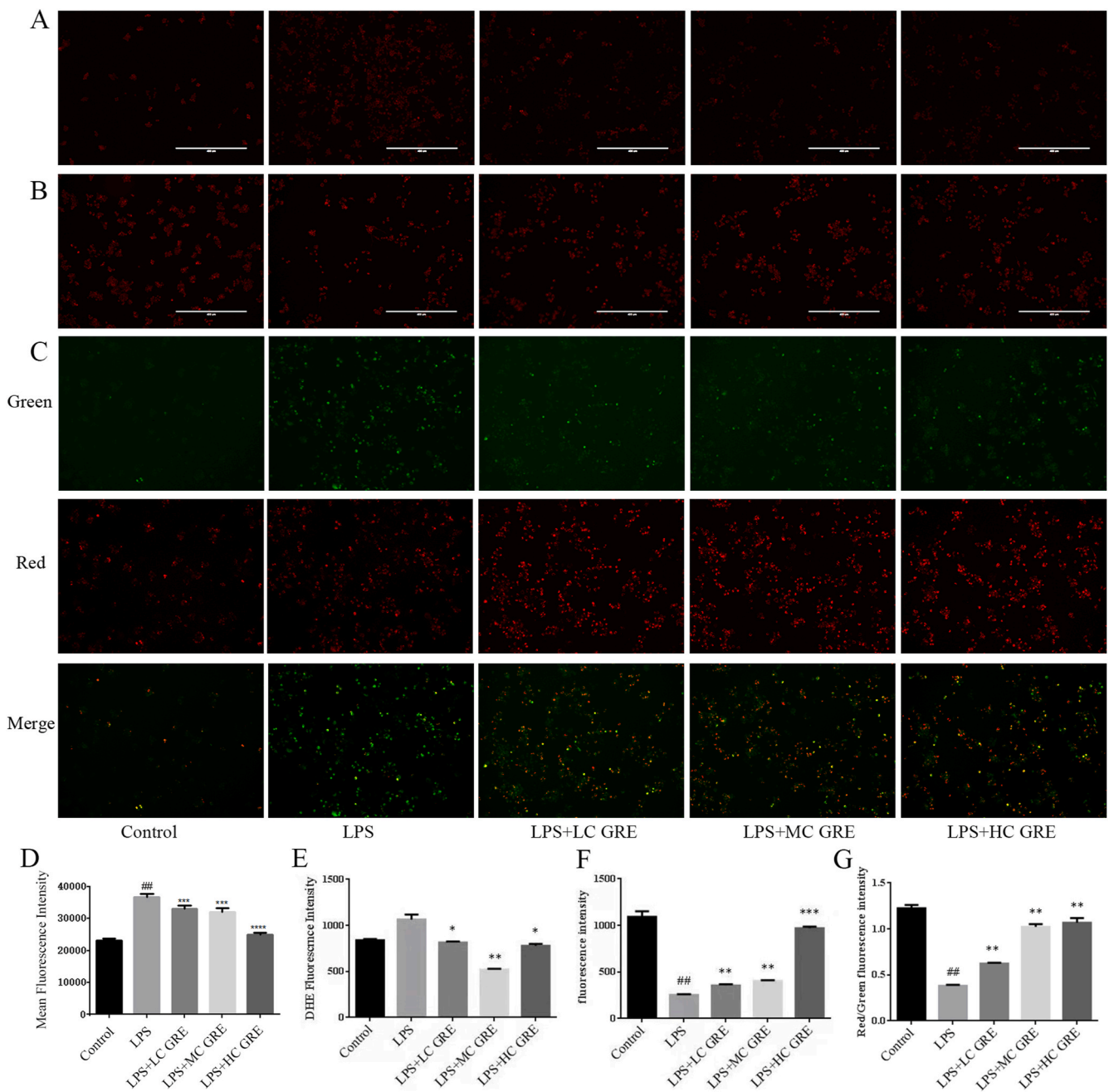


Fig. 4. Ginseng root extract (GRE) attenuates LPS-induced oxidative damage and mitochondrial dysfunction in RAW264.7 cells. (A) DHE staining for O₂·, (B) Mito-tracker staining for mitochondrial number, (C) JC-1 staining for MMPs, and (D) DCFH-DA staining for ROS. (E) The quantitative data of panel (A). (F) The quantitative data of panel (B). (G) The quantitative data of panel (C).

ZORBAX SB-Aq HPLC column (5 μ m, 4.6 $\times$ 250 mm) was used for ginsenoside separation. The temperature of the column was maintained at 27 $^{\circ}$ C. The flow rate was 1 mL/min. The mobile phase consisted of solvent A (acetonitrile) and solvent B (0.05% phosphate buffer). The gradient elution program was as follows: 0–30 min, A 19%; 30–40 min, A 19%–29%; 40–70 min, A 29%; 70–80 min, A 29%–32%; 80–90 min, A 32%–42%; 90–105 min, A 42%–60%; 105–113 min, A 60%. The flow rate was set at 1 mL/min. The quantitative method was the external standard method.

2.5. Cell culture and LPS treatment

RAW264.7 (ATCC, TIB-71) cells were cultured in Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum

(FBS) at 37 $^{\circ}$ C in a 5% CO₂ atmosphere. After overnight culture in 6-well plates, RAW264.7 cells (2 $\times$ 10⁵ cells/well) were treated with 1 μ g/mL LPS or GRE/LPS for 24 h.

2.6. Small interfering RNA (siRNA) transfection

SiRNAs were transiently transfected into RAW264.7 cells using the Advanced DNA RNA transfection reagent. The p62 siRNA sequences were 5'-GCACAGAAGACAAGAGUAATT-3' and 5'-UUACUCUUGUCUU-CUGUGCTT-3'. The control siRNA sequences were 5'-ACGUGA-CACGUUCGGAGAATT-3' and 5'-UUCUCCGAACGUGUCACGUTT-3'. All siRNAs were transfected at 30 nM concentrations. After a 24-h incubation period, cells were treated with LPS or GRE/LPS for an additional 24 h and then analyzed with downstream assays.

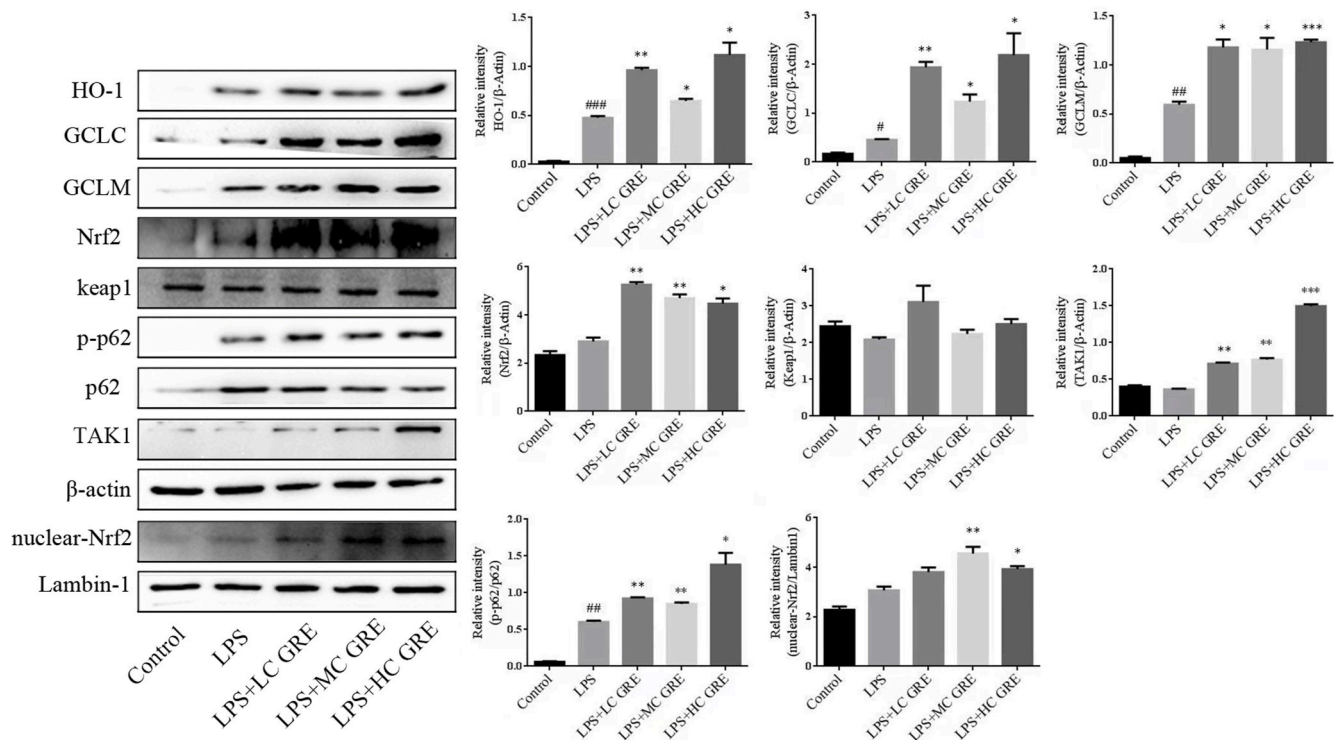


Fig. 5. Ginseng root extract (GRE) attenuates LPS-triggered oxidative damage via the p62-Nrf2-Keap1 pathway. Cells were treated with LPS or LPS/GRE for 24 h. The protein expression levels of HO-1, GCLC, GCLM, Nrf2, Keap1, β-actin, as well as p62, p-p62, and TAK1, were measured by western blotting. Nuclear fractions were extracted and examined by western blotting. β-actin and Lamin B1 protein levels were detected to confirm equal protein amounts in the cytoplasmic fractions and nuclear extracts, respectively.

2.7. Fluorescent staining

The following fluorescent probes were used in this study: O2⁻ (DHE, 10 μM) formations, mitochondria numbers (Mitotracker RedCMXRos, 200 nM), mitochondrial membrane potential (JC-1, 1×), ROS (DCFH-DA, 10 μM) and NO (DAF-FM DA, 5 μM). Cells were incubated in culture medium containing the fluorescent probe for 20–30 min at 37 °C. Next, the plates were washed three times with culture medium (FBS-free) or staining buffer, and then observed with a fluorescence microscope (Thermo Fisher Scientific, Waltham, MA, USA). Densitometry analysis was performed with Graphpad Prism 6.0 software. For detecting intracellular NO and ROS, a microplate reader was used to measure absorbance of cells at 450 nm (Tecan Infinite 200PRO, Tecan, Mannedorf, Switzerland).

2.8. Transmission electron microscopy (TEM)

After treatment, cells were harvested and fixed in glutaraldehyde (4%) overnight at 4 °C. After dehydration, ultrathin sections were prepared and stained with uranyl acetate/lead citrate. Images were captured using a transmission electron microscope (H-7650, Hitachi, Japan).

2.9. qRT-PCR

Cells (2×10⁵ cells/well) were treated with LPS or GRE/LPS for 24 h. After centrifugation (800×g; 5 min; 25 °C), the cells were harvested and washed three times in PBS. Total RNA was extracted with TRIzol and reverse transcribed into cDNA using the TAKARA reverse transcription kit. qRT-PCR was performed with the SYBR Premix Ex Taq kit and the CFX Connect™ Real-Time system (Bio-Rad Laboratories, Inc.). The relevant PCR primers are shown in Table 3. The thermal cycling conditions were as follows: 95 °C for 3 min; 40 cycles of 95 °C for 10 s and

55 °C for 30 s; 95 °C for 10 s; 65 °C for 5 s and 95 °C for 5 s. Data were calculated relative to GAPDH expression levels according to the 2^{-ΔΔCq} method.

2.10. Western blotting

Cells were harvested by centrifugation (800×g; 5 min; 25 °C) and resuspended in RIPA total protein extraction lysis buffer. A BCA assay was used to detect the protein concentrations. Loading buffer was added to the samples, and they were boiled for 15 min. Proteins were separated by SDS-PAGE on a 12% gel, then transferred to a PVDF membrane. The membranes were blocked with 5% dry non-fat milk (BD Biosciences) at room temperature for 0.5–1 h. After washing in PBST three times, the membranes were incubated with the specific primary antibodies at 1:1000 dilutions overnight at 4 °C. They were then washed in PBST three times and incubated with the appropriate secondary antibody (1:1000 dilution) for 1 h at room temperature. Protein bands were visualized using the Ultra High Sensitivity ECL Substrate kit. Immunoreactivity was visualized by chemiluminescence, and densitometric analysis was performed with ImageJ Software (v1.8.0).

2.11. Measurements of TNF-α and IL-6

After treatment with LPS or GRE/LPS, cell supernatants were collected. The TNF-α and IL-6 concentrations of the supernatants were estimated using ELISA kits.

2.12. DSS-induced colitis

Colitis was induced in mice using DSS with a molecular weight of 36,000–50,000 (MP Biomedicals, Santa Ana, CA, USA). C57BL/6J male mice (18–25 g, 8 weeks old) were pre-protected with the GRE and positive drugs sulfasalazine for 3 days, during which they were given

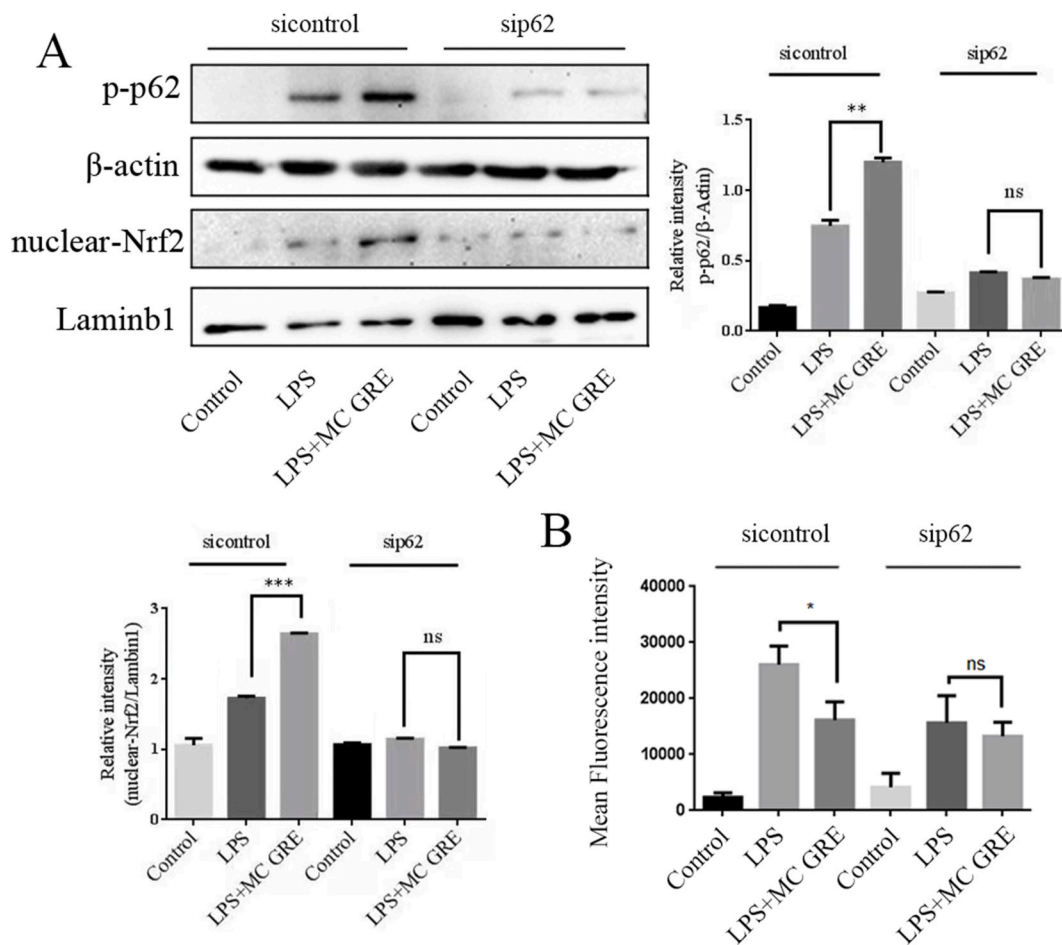


Fig. 6. Knockdown of p62 decreases the anti-oxidative function of Ginseng root extract (GRE). (A) After cells were treated with LPS or LPS/GRE and transfected with si-p62 or si-control, the protein expression levels of p-p62, β -actin, nuclear-Nrf2, and Lamin B1 were determined by western blotting. (B) Production of intracellular ROS was detected after cells were treated with LPS or LPS/GRE and transfected with si-p62 or si-control.

normal drinking water. The dose of GRE and sulfasalazine were 0.18 g/kg and 0.3 g/kg, the perfusion volume of each mouse was 0.2 mL. Then, 3% DSS solution was provided to mice in drinking water, the dose and volume of GRE and sulfasalazine remained the same as before. The animals were sacrificed after 10 days.

2.13. Tissue collection

From each group, the mouse colon tissues were collected and fixed in 4% paraformaldehyde fixing solution. The paraffin-embedded tissue was subjected to hematoxylin and eosin (H&E) staining using a standard procedure. Tissues were also collected and treated for further western blot and qRT-PCR analyses.

2.14. Statistical analyses

All data represent the mean $\pm$ SEM of at least three independent experiments. Statistical comparisons were made by one-way analysis of variance (ANOVA) and Dunnett's post-hoc test. * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; not significant (ns) $p > 0.05$.

3. Results

3.1. Extract analysis

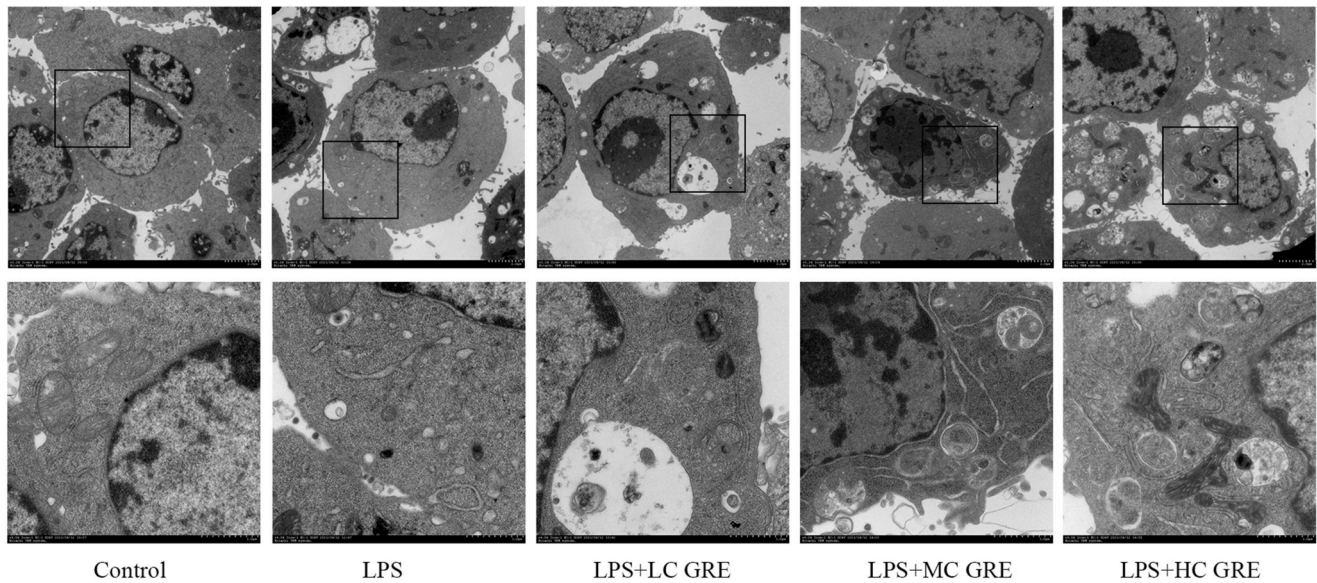
The GRE was weighed and prepared into 500 mg/mL liquid storage. The storage was then filtered with 0.22 μ m filters and diluted to three

concentrations: high concentration (HC) 500 μ g/mL, middle concentration (MC) 125 μ g/mL, and low concentration (LC) 32.5 μ g/mL. Through composition analysis, HC extracts contained 21.25 μ g/mL total ginsenoside, 14.75 μ g/mL total polysaccharide, and 36 μ g/mL total protein. MC extracts contained 5.32 μ g/mL total ginsenoside, 3.69 μ g/mL total polysaccharide, and 9 μ g/mL total protein. LC extracts contained 1.33 μ g/mL total ginsenoside, 0.92 μ g/mL total polysaccharide, and 2.25 μ g/mL total protein (Table 1). The total ginsenoside in GRE was analyzed by HPLC. The chromatograms, including standard and sample chromatograms, are shown in Fig. S1. The main components and contents of TGN are listed in Table 2.

3.2. GRE inhibits inflammation and pro-inflammatory cytokines induced by LPS treatment

RAW264.7 cells were treated with LPS or GRE/LPS to examine the effect of GRE on LPS-induced inflammation. Firstly, the intracellular NO concentration was measured. As shown in Fig. 1A, GRE treatment significantly reduced the NO level in the GRE/LPS group compared with that of the LPS alone group. Pro-inflammatory cytokines play important roles in inflammatory processes. To further study the anti-inflammatory effects of GRE, the levels of pro-inflammatory cytokines in the culture medium were detected using ELISAs. As shown in Fig. 1B, LPS strongly activated the release of TNF- α and IL-6, while secreted levels of these cytokines were notably decreased in cells treated with GRE. Additionally, qRT-PCR was performed to detect the transcription levels of cytokines IL-10, IL-1 β , and IL-6. IL-1 β and IL-6 are considered classic pro-

A



B

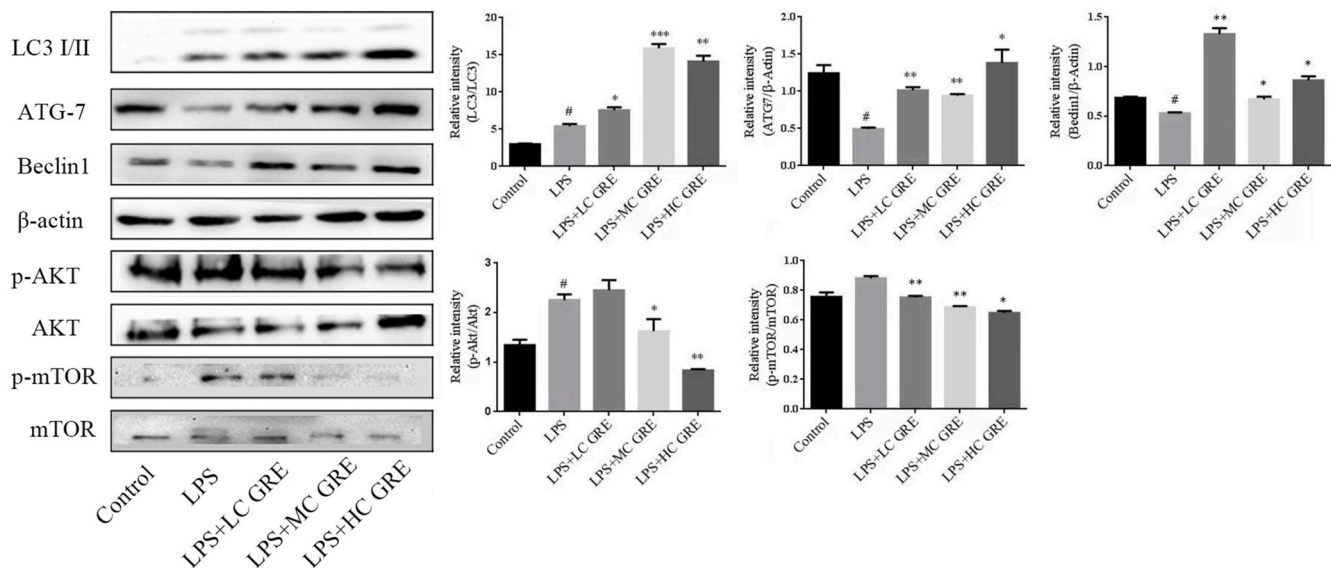


Fig. 7. Ginseng root extract (GRE) protects RAW264.7 cells from inflammatory effects by activating autophagy. (A) After being treated with LPS or LPS/GRE for 24 h, cells were fixed, sectioned, and observed by transmission electron microscopy. (B) Protein expression levels of LC3 I/II, ATG-7, Beclin1, p-mTOR, mTOR, p-AKT, AKT, and β-actin were detected by western blotting.

inflammatory cytokines that recruit immune cells to the location of inflammation. While IL-10 is an anti-inflammatory cytokine, it can control release of pro-inflammatory cytokines induced by LPS (Hedrich, 2016). As expected, treatment with GRE led to a remarkable suppression of IL-1β and IL-6 mRNA levels, but increased transcription of IL-10 (Fig. 1C).

3.3. GRE blocks LPS-triggered activation through the NF-κB and MAPK signaling pathways

NF-κB is a crucial factor involved in the regulation of pro-inflammatory and anti-inflammatory mediators. The expression levels of COX-2 and iNOS were enhanced by activation of NF-κB (Surh et al., 2001). To determine whether GRE can inhibit LPS-triggered NF-κB activation, western blotting was employed to examine the protein

expression levels of COX-2, iNOS, and NF-κB p65. The results demonstrated that LPS treatment significantly activated the nuclear translocation of NF-κB and increased the expression of both COX-2 and iNOS. However, GRE treatment effectively inhibited LPS-induced iNOS and COX-2 upregulation, as well as the nuclear translocation of NF-κB p65 (Fig. 2). MAPKs also play critical roles in regulating cellular responses to cytokines. The phosphorylation of MAPKs, which is activated by LPS, is associated with the activation of COX-2 and iNOS (Liu et al., 2014). Therefore, we measured the inhibitory activities of GRE on the phosphorylation of MAPKs induced by LPS. The results suggest that LPS treatment could indeed stimulate phosphorylation of MAPKs, while GRE treatment reduced this phosphorylation (Fig. 2). Similarly, asperuloside inhibited LPS-induced nuclear translocation of NF-κB p65 and COX2 upregulation (Fig. 3A). SP600125 inhibited LPS-induced phosphorylation of JNK, nuclear translocation of NF-κB p65, and COX-2 upregulation

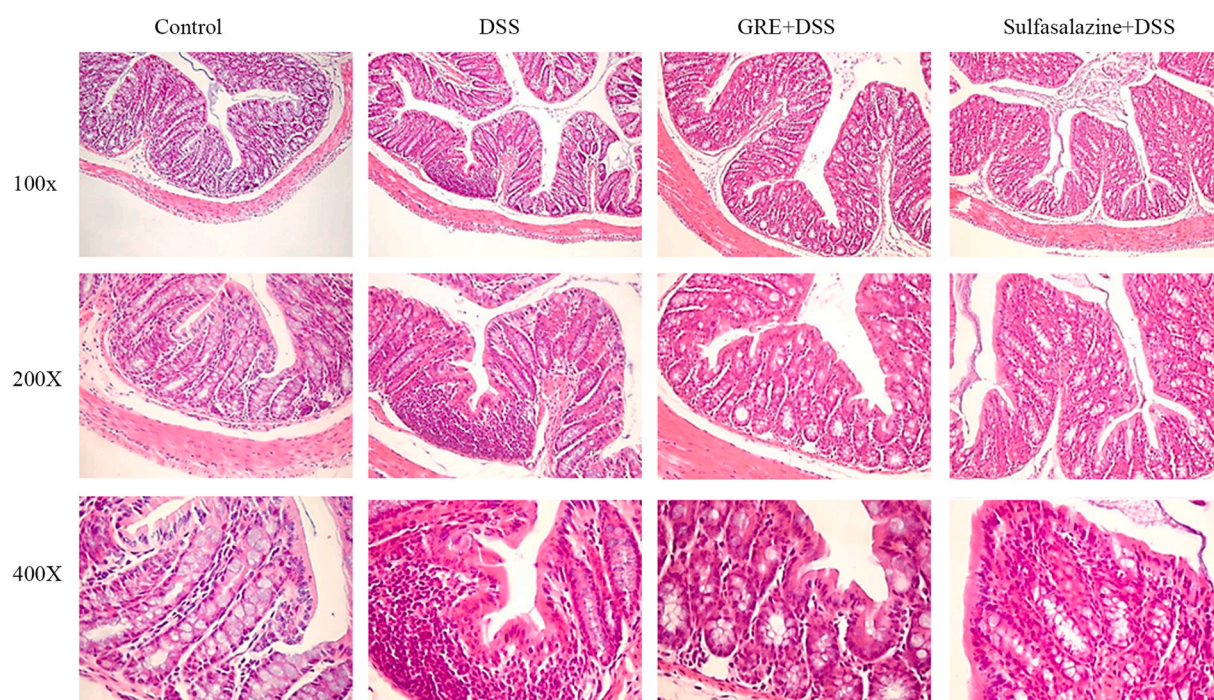


Fig. 8. Ginseng root extract (GRE) treatment significantly relieves colon injury induced by DSS. Representative H&E sections of colon, Tissue morphology was analyzed after DSS challenge.

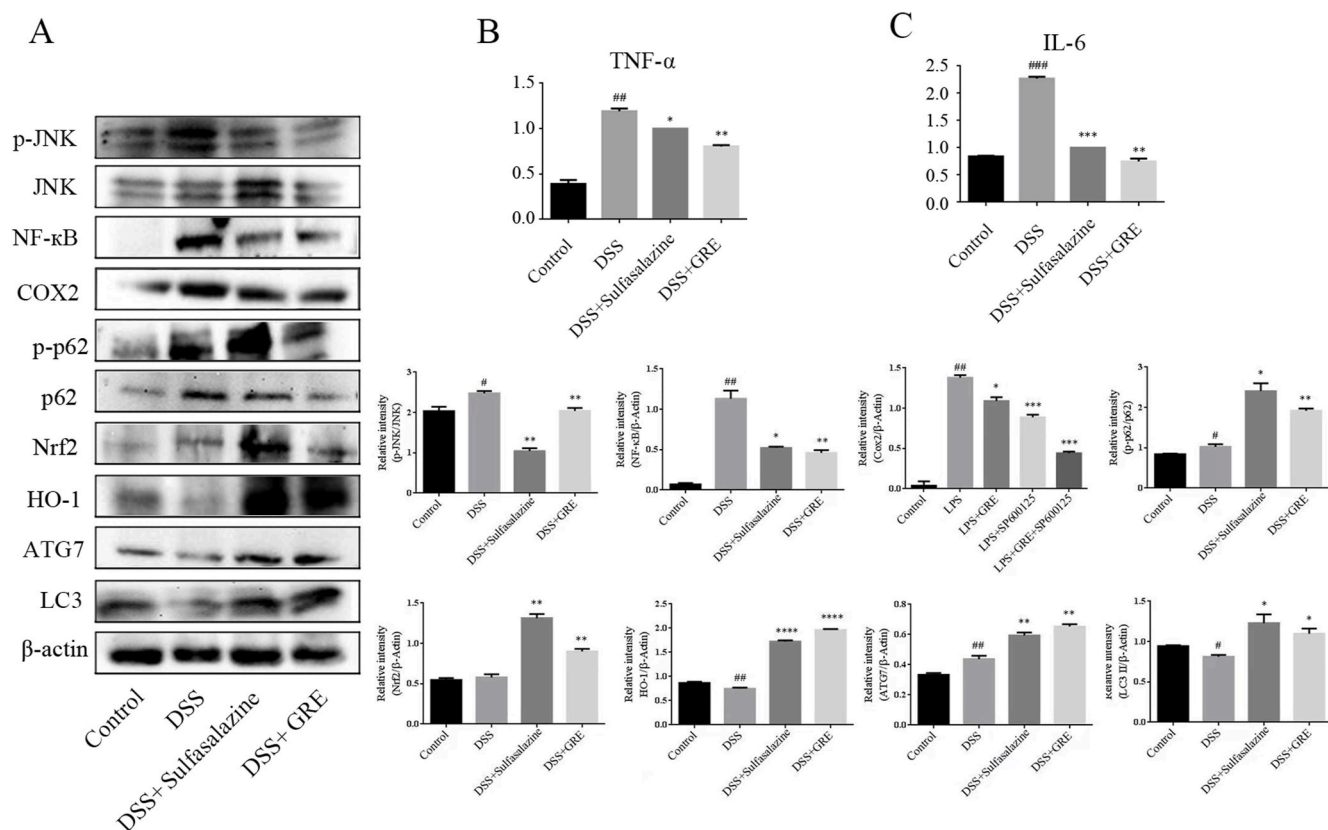


Fig. 9. Ginseng root extract (GRE) exerts both anti-inflammatory and anti-oxidative functions by targeting the MAPK/NF- κ B and p62-Nrf2-Keap1 pathways, as well as the autophagy process, in DSS-induced colitis. (A) The protein expression levels of HO-1, Nrf2, p62, p-p62, p-JNK, JNK, NF- κ B, COX-2, β -actin, as well as LC3 and ATG7, were measured by western blotting. The mRNA levels of (B) TNF- α and (C) IL-6 were detected by qRT-PCR.

(Fig. 3B). These data demonstrate that GRE can relieve LPS-induced inflammation via regulating NF- κ B and MAPK signaling.

3.4. GRE decreases LPS-triggered oxidative damage and mitochondrial dysfunction

LPS has been previously shown to induce excessive ROS production implicated in the acceleration of inflammation development from further induction of cytokine expression (Gill et al., 2010). To assess the ability of GRE to relieve the levels of ROS generated by LPS, we analyzed the production of ROS, as well as the O_2^- level. As depicted in Fig. 4A, D, and E, LPS significantly enhanced the intracellular ROS and O_2^- levels. As expected, GRE treatment remarkably decreased both ROS and O_2^- formation. Mitochondria were the main locations of ROS generation, and oxidative stress was associated with mitochondrial dysfunction, including decreased mitochondrial numbers and an imbalance of mitochondria membrane potential (MMP) (Ryan et al., 2015). Mitochondrial numbers in cells were reduced by LPS treatment, while GRE rescued this damage (Fig. 4B and F). Furthermore, MMP was measured by JC-1 fluorescence probes. When the MMP was high, JC-1 aggregated in the matrix of mitochondria and formed polymers, then produced red fluorescence. When the MMP was low, JC-1 remained as a monomer. This resulted in a production of green fluorescence. The proportion of mitochondrial depolarization is often measured by the relative ratio of red and green fluorescence. Fig. 4C and G shows that the ratio of red and green fluorescence in cells was notably declined after LPS treatment, while this ratio decrease was prevented with GRE treatment.

3.5. GRE attenuates LPS-triggered oxidative injury through the p62-Nrf2-Keap1 pathway

Nrf2 is considered to be an important factor for protecting cells against excessive ROS during the inflammatory response. TGF- β -activated kinase 1 (TAK1) is a regulator of p62-dependent Keap1 degradation and sustains the baseline level of Nrf2 (Hashimoto et al., 2016). Hence, we examined the effect of GRE on the regulation of the p62-Nrf2-Keap1 pathway proteins in LPS-induced inflammation. Our results suggest that GRE could significantly promote both Nrf2 expression and its translocation from the cytoplasm to the nucleus. However, there was no effect on Keap1 expression. We detected the protein expression levels of Nrf2 downstream factors HO-1, GCLC, and GCLM using western blots and found that GRE treatment significantly promoted expression of these proteins (Fig. 5). Additionally, GRE could induce the phosphorylation of p62 and promote TAK1 expression. To further research the effect of p62 on the anti-oxidative function of GRE, we designed siRNAs to specifically target and knock down p62 mRNA. Fig. 6A and B show that si-p62 treatment considerably inhibited the anti-oxidative activity of GRE, including the regulation of Nrf2 nuclear translocation and the ROS level. These data suggest that GRE can attenuate LPS-triggered ROS via activation of Nrf2 by enhancing phosphorylation of p62.

3.6. GRE activates autophagy in RAW264.7 cells to reduce inflammatory effects

Cells suffer excessive oxidative stress during inflammatory responses, and autophagy has been demonstrated to help reduce these effects. To measure the autophagy level, TEM was used to survey the accumulation of autophagosomes in LPS and LPS/GRE-treated cells. The results exhibited an increased number of autophagosomes compared with that of the control group (Fig. 7A). We then detected the expression levels of some key autophagy-related proteins. Western blot analyses suggest that LC3 II, ATG-7, and Beclin-1 protein levels were enhanced after GRE treatment in RAW264.7 cells compared with those of the LPS-treated group. Furthermore, the AKT-mTOR pathway is associated with regulating cell inflammation and autophagy. As shown in Fig. 7B, GRE

treatment induced downregulation of p-mTOR/mTOR and p-AKT/AKT. These data demonstrate that GRE can promote autophagy through the AKT-mTOR pathway in RAW264.7 cells.

3.7. GRE alleviates DSS-induced colitis through reduced inflammatory and oxidative stress

To investigate the effects of GRE on DSS-induced colitis in a mouse model, colon morphology was observed for each group using H&E staining. The DSS-challenged mice showed remarkable colon injury, which was characterized by a reduced number of epithelial goblet cells and increased inflammatory cell infiltration between the intestinal glands. Meanwhile, similar to the sulfasalazine positive control, GRE treatment significantly relieved colon injury (Fig. 8). To determine the specific mechanism of GRE-eased DSS-induced colitis, western blotting and qRT-PCR were respectively employed to examine the protein and mRNA expression levels of relevant inflammatory cytokines and factors. As shown in Fig. 9A, GRE treatment decreased the phosphorylation of JNK and the expression levels of NF- κ B and COX-2. However, it increased the phosphorylation of p62 and the expression levels of Nrf2, HO-1, GCLC, LC3 II, and ATG7. Additionally, treatment with GRE led to a remarkable suppression of DSS-induced TNF- α and IL-6 production (Fig. 9B and C). These results suggest that GRE can exert both anti-inflammatory and anti-oxidative functions by targeting the MAPK/NF- κ B and p62-Nrf2-Keap1 pathways, as well as the autophagy process, in a DSS-induced colitis mouse model.

4. Discussion

Herbal medicine has become increasingly popular in current society. Ginseng is known as the “king of herbs,” as it has been used for thousands of years. Ginseng roots contain ginsenosides, polysaccharides, proteins, phenols, polyacetylenes, sesquiterpenes, and alkaloids (Cambria et al., 2021). Various components of ginseng have a wide range of pharmacological effects, including anti-inflammatory and antioxidant properties. Ginsenoside Rg1 protects ischemia reperfusion-induced nerve injury through the miR-144/Nrf2/ARE axis (Chu et al., 2019). Ginsenoside Rd inhibits ROS products induced by high glucose in mesangial cells and prevents renal insufficiency in db/db mice (Jung et al., 2021). In transient focal ischemic encephalopathy, ginsenoside Rd reduces inflammation by inhibiting the expression of iNOS and COX-2 (Kim et al., 2013). The radiation protective effect of acidic ginseng polysaccharide on mice is related to the regulation of antioxidant enzymes (Sun, 2011). Ginseng fruit polysaccharides can inhibit T cell differentiation, promote apoptosis, and enhance the effect of 5-fluorouracil on inflammatory colon cancer (Wang et al., 2020). Ginseng total protein plays a protective role in mitochondrial dysfunction and neurodegeneration in drosophila models of Parkinson's disease by inducing mitochondrial unfolded protein response (Liu et al., 2020). Ginseng oligopeptide inhibits edema by regulating the MAPK and NF- κ B pathways, and also improves the inflammatory response induced by dextran in model rats (Xu et al., 2019). In this study, we detected the anti-oxidative and anti-inflammatory functions of total GRE. These could be a synergistic effect of different components of ginseng root.

In traditional Chinese medicine, ginseng has been used to treat diverse metabolic-related diseases, like diabetes, in the clinic (Zhang et al., 2021). Long-term, chronic inflammation contributes to several metabolic diseases including obesity, diabetes, and atherosclerosis. Here, we investigated the anti-oxidative and anti-inflammatory functions of GRE in LPS-induced RAW264.7 cells. We found that GRE could markedly decrease NO release, as well as the production of pro-inflammatory cytokines via the NF- κ B and MAPK pathways in these cells. MAPKs, including JNK, ERK, and p38, are key kinases of anti-inflammatory signaling in macrophages. The activation of these kinases requires that they be phosphorylated. Our results show that GRE can significantly reduce the phosphorylation of JNK, ERK, and p38 that

was activated by LPS. Additionally, plasticity and diversity are two hallmarks of macrophages. M1 macrophages are associated with a pro-inflammatory microenvironment, while M2 macrophages are involved in responses to anti-inflammatory reactions and tissue remodeling (Shapouri-Moghaddam et al., 2018). In our results, GRE treatment promoted the transcription of anti-inflammatory cytokine IL-10 (M2 macrophage-related), but decreased the transcriptional levels of pro-inflammatory cytokines IL-1 β and IL-6 (M1 macrophage-related). These data imply that the pharmacological mechanism of GRE may be related to macrophage polarization.

p62 is considered to be a multifunctional adaptor protein associated with numerous cellular processes (autophagy, cell death and survival, DNA repair, oxidative stress, and inflammation). Phosphorylation of p62 at serine 351 allows for a direct interaction with the Nrf2 binding domain on Keap1 (349-DPSTGE-354). Then, p-p62 can recruit Keap1 into autophagosomes, where it is degraded through autophagy. p62 is also a transcriptional target of Nrf2. Nrf2 translocates into the nucleus and promotes p62 transcription, which in turn results in a second round of Nrf2 activation and forms a positive feedback loop between Nrf2 and p62 (Fan et al., 2018). In our study, GRE significantly stimulated phosphorylation of p62, and also facilitated the activation of Nrf2 and the expression of downstream factors. However, we did not observe any clear change in Keap1 expression. This may be because the level of p62-induced Keap1 degradation is similar to Keap1 degradation by the proteasome pathway. Moreover, TEM images and western blot analysis revealed that GRE treatment could induce autophagy and increase the expression of autophagy-related proteins. These results demonstrate that GRE can attenuate LPS-triggered inflammation in RAW264.7 cells by regulating the NF- κ B, MAPK, and p62-Keap1-Nrf2 signaling pathways, as well as the autophagy process.

IBD is characterized by chronic relapsing inflammation with intestinal epithelial injury and immune homeostasis disruption. Recent studies have shown that ginseng has a specific therapeutic effect on IBD through cell and animal model experiments. Red ginseng improves the intestinal microbial structure and relieves the symptoms of UC (Guo et al., 2015). Fermenting wild ginseng reduces the severity of colitis and macrophage infiltration in colon tissue, and its mechanism is related to inhibition of the NF- κ B signaling pathway (Triantafyllidis et al., 2016). In this study, GRE was used to treat lab mice with DSS-induced colitis. Our results suggest that GRE treatment remarkably rescued the colitis-associated reduction of epithelial goblet cells and increase in inflammatory cell infiltration between the intestinal glands. Overall, GRE treatment could ease inflammation in DSS-induced colitis via the NF- κ B, MAPK, and p62-Nrf2 signaling pathways, and the autophagy process.

5. Conclusions

Here, we have demonstrated that GRE has strong anti-inflammatory and anti-oxidative functions in LPS-treated RAW264.7 cells and DSS-induced colitis. These data provide novel insights into the protective effects of GRE. These data connect the anti-inflammatory and antioxidant functions of ginseng to the regulation of autophagy. This research confirms the pharmacological activity of ginseng, lays a foundation for the development of ginseng-based products, and provides theoretical support for the clinical application of ginseng in inflammation-related diseases. However, more research is needed to identify the specific functional components in GRE.

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CRediT authorship contribution statement

Song Yang: Investigation, Formal analysis. **Fangyu Li:** Investigation, Formal analysis. **Shuyan Lu:** Investigation. **Limei Ren:** Validation. **Shuai Bian:** Validation. **Meichen Liu:** Writing – review & editing. **Daqing Zhao:** Funding acquisition. **Siming Wang:** Conceptualization, Writing – original draft. **Jiawen Wang:** Conceptualization, Methodology, Writing – original draft.

Declaration of competing interest

The authors declare that they have no known competing financial interests of personal relationships that could have appeared to influence the work reported in this paper.

Abbreviations

AKT	AKT serine/threonine kinase
AREs	antioxidant response elements
COX-2	cyclooxygenase-2
DHE	dihydroethidium
DSS	dextran sulfate sodium
ELISA	enzyme-linked immunosorbent assay
ERK	extracellular signal-related kinase
FoxO1	forkhead box O1
GCLC	glutamate-cysteine ligase catalytic subunit
GCLM	glutamate-cysteine ligase modifier subunit
GRE	ginseng root extract
HC	high concentration
HMGB1	high mobility group box 1
HO-1	oxygenase 1
IBD	inflammatory bowel disease
IL-6	interleukin-6
iNOS	nitric oxide synthase
JNK	c-Jun NH2-terminal kinase
Keap1	kelch-like ECH-associated protein1
LC	low concentration
LPS	lipopolysaccharide;
MAPKs	mitogen-activated protein kinases
MC	middle concentration
mTOR	mechanistic target of rapamycin kinase
NF- κ B	nuclear factor kappa B subunit 1
NO	Nitric oxide;
Nrf2	nuclear factor erythroid 2-related factor 2
qRT-PCR	quantitative real-time polymerase chain reaction
ROS/RNS	reactive oxygen/nitrogen species
TAK1	TGF- β -activated kinase 1
TEM	transmission electron microscopy
TGF- β 1	transforming growth factor- β 1
TNF- α	tumor necrosis factor- α ;
UV	ultraviolet.

Appendix A. Supplementary data

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